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NEET · PHYSICS · CHAPTER-WISE PYQS

Dual Nature of Radiation and Matter

NEET Physics · Previous Year Questions with Answer Key

87	37	2000 – 2026	1	English
QUESTIONS	PAPERS COVERED	YEARS	OUT OF SYLLABUS	LANGUAGE

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MCQ (Single Correct Answer)	85 questions
MCQ (More than One Correct Answer)	1 question
Out of syllabus	1 question
Answer key	at the end

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MCQ (Single Correct Answer) 85 questions

Q1. RE-NEET 2026

A photon and an electron, each of 20 eV energy, move in free space. The ratio of linear momentum of electron p_e to that of photon p_{Ph}, $\frac{p_e}{p_{Ph}}$ is:

[Take speed of light $= 3 \times 10^8\ ms^{-1}$, charge of electron $= -1.6 \times 10^{-19} C$ and mass of electron $= 9 \times 10^{-31}\ kg$]

Open solution

(A) 275 (B) $\frac{2}{450}$ (C) $\frac{1}{250}$ (D) 225

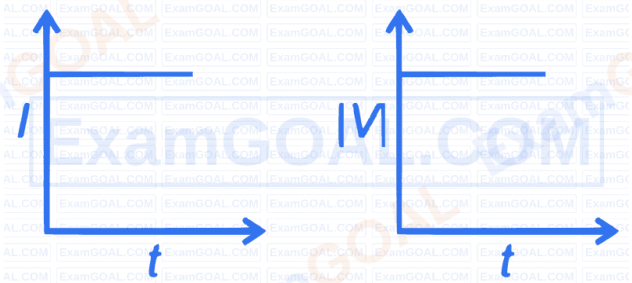
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Q2. RE-NEET 2026

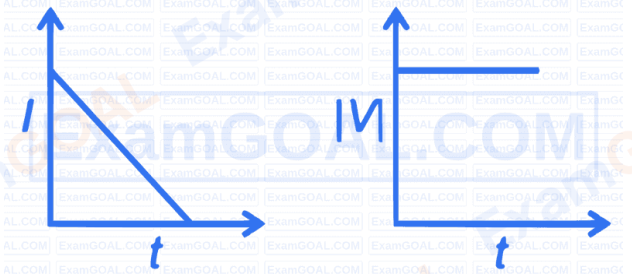
A beam of light falls on a metal surface such that photo-electrons are generated. If power of the light source starts to decrease linearly with time t, then variation of the photocurrent I and magnitude of the stopping potential $|V|$ with time is best represented by:

Open solution

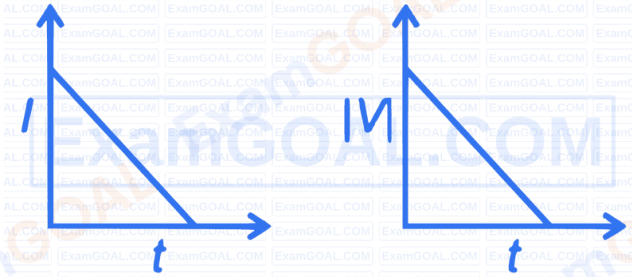
(A)



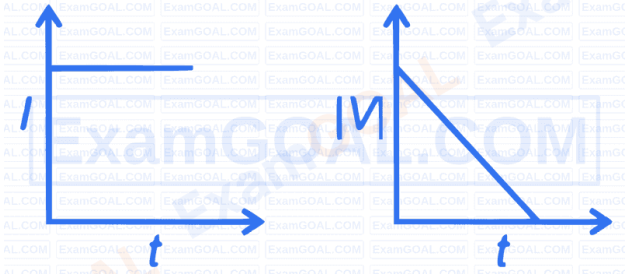
(B)



(C)



(D)



Q3. RE-NEET 2026

A ray of light with wavelength λ is incident on three different photoelectric cells namely 1,2 and 3 . The threshold wavelength of these photoelectric cells are λ_1, λ_2, and λ_3, respectively and the magnitude of stopping potentials of these cells are V_1, V_2 and V_3, respectively. The relation between λ and threshold wavelengths are $\lambda_1 < \lambda, \lambda_2 > \lambda$ and $\lambda_3 \gg \lambda$. The correct option is:

Open solution

(A) $V_1 < V_2, V_3 = 0$ (B) $V_1 = 0, V_2 < V_3$ (C) $V_1 = 0, V_2 > V_3$ (D) $V_1 > V_2, V_3 = 0$

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Q4. NEET 2026

Open solution

Match List I with List II.

List-I		List-II	
A.	$E = hv$	I.	de Broglie wavelength
B.	Diffraction and Interference	II.	Particle nature of light
C.	$\lambda = h/p$	III.	Wave nature of light
D.	Compton effect	IV.	Energy of photon

(A) A-IV, B-I, C-II, D-III
(B) A-IV, B-III, C-II, D-I
(C) A-I, B-IV, C-III, D-II
(D) A-IV, B-III, C-I, D-II

Q5. NEET 2026

Open solution

For a metal of work function 6.6 eV , which of the following wavelengths of incident radiation does not give rise to the photoelectric effect?

(Take Planck's constant as $6.6 \times 10^{-34}\ \mathrm{J\ s}$)

(A) 100 nm **(B)** 150 nm **(C)** 200 nm **(D)** 50 nm

Q6. NEET 2025

Open solution

De-Broglie wavelength of an electron orbiting in the $n = 2$ state of hydrogen atom is close to

(Given Bohr radius $= 0.052\ \mathrm{nm}$)

(A) 1.67 nm **(B)** 2.67 nm **(C)** 0.067 nm **(D)** 0.67 nm

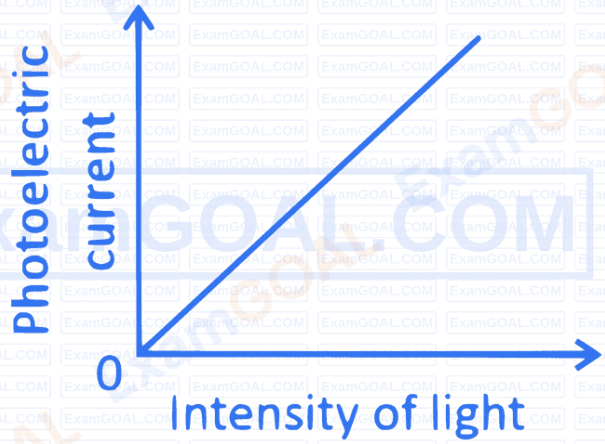
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Q7. NEET 2025

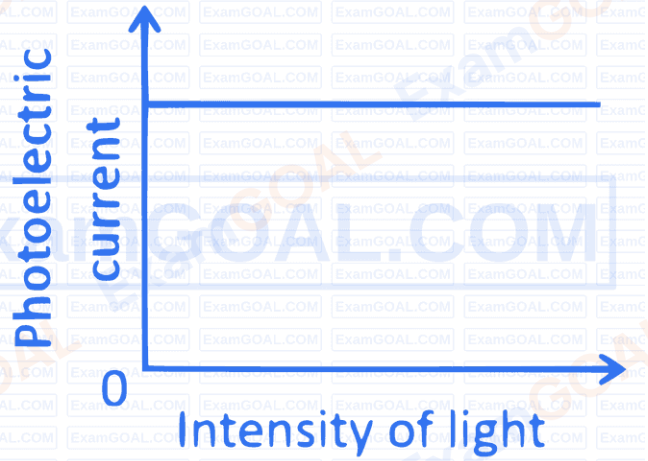
Which of the following options represent the variation of photoelectric current with property of light shown on the x-axis?

Open solution

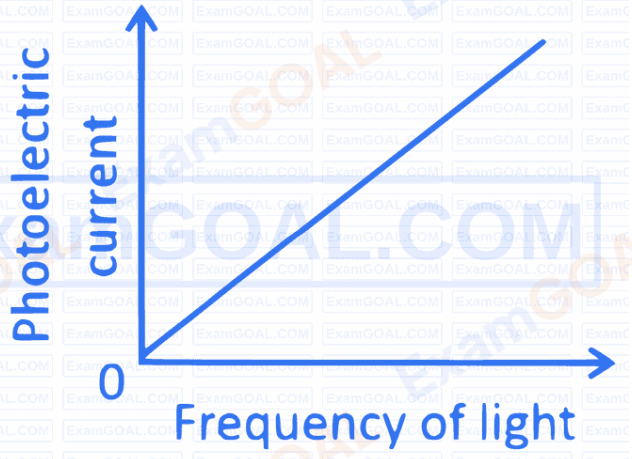
A.



B.

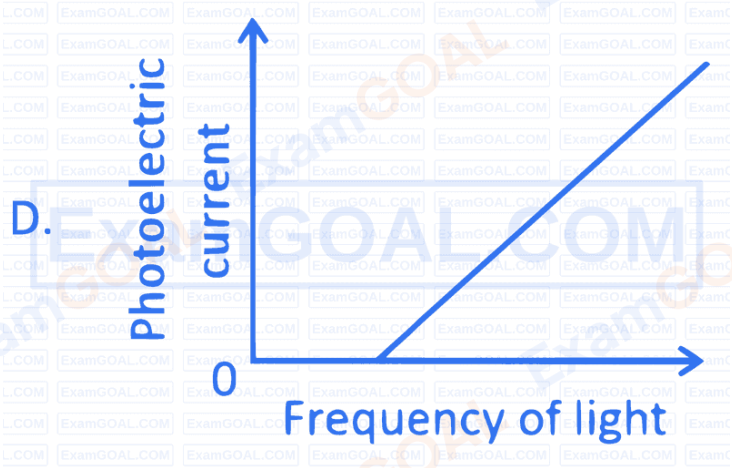


C.



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D.



(A) A and D (B) B and D (C) A only (D) A and C

Q8. NEET 2025

A photon and an electron (mass m) have the same energy E. The ratio ($\lambda_{\text{photon}} / \lambda_{\text{electron}}$) of their de Broglie wavelengths is: (c is the speed of light)

(A) $c\sqrt{\frac{2m}{E}}$ (B) $\frac{1}{c}\sqrt{E/2m}$ (C) $\sqrt{E/2m}$ (D) $c\sqrt{2mE}$

Open solution

Q9. NEET 2024 (Re-Examination)

An electron and an alpha particle are accelerated by the same potential difference. Let λ_e and λ_α denote the de-Broglie wavelengths of the electron and the alpha particle, respectively, then:

(A) $\lambda_e > \lambda_\alpha$ (B) $\lambda_e = 4\lambda_\alpha$ (C) $\lambda_e = \lambda_\alpha$ (D) $\lambda_e < \lambda_\alpha$

Open solution

Q10. NEET 2024 (Re-Examination)

If ϕ is the work function of photosensitive material in eV and light of wavelength of numerical value $\lambda = \frac{hc}{e}$ metre, is incident on it with energy above its threshold value at an instant then the maximum kinetic energy of the photo-electron ejected by it at that instant (Take h-Plank's constant, c-velocity of light in free space) is (in SI units):

(A) $e + 2\phi$ (B) $2e - \phi$ (C) $e - \phi$ (D) $e + \phi$

Open solution

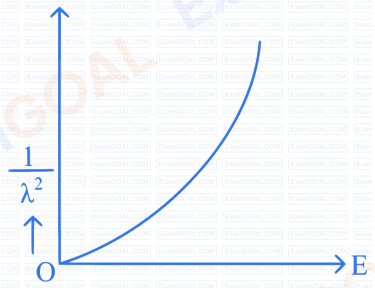
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Q11. NEET 2024

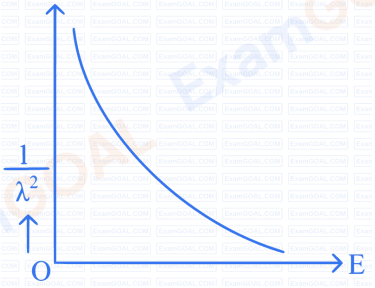
The graph which shows the variation of $\left(\frac{1}{\lambda^2}\right)$ and its kinetic energy, E is (where λ is de Broglie wavelength of a free particle):

Open solution

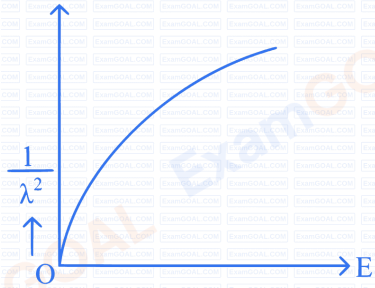
(A)



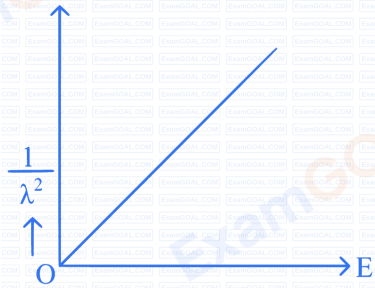
(B)



(C)



(D)



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Q12. NEET 2024

If c is the velocity of light in free space, the correct statements about photon among the following are:

A. The energy of a photon is $E = hv$.

B. The velocity of a photon is c.

C. The momentum of a photon, $p = \frac{hv}{c}$.

D. In a photon-electron collision, both total energy and total momentum are conserved.

E. Photon possesses positive charge.

Choose the correct answer from the options given below:

(A) A and B only
(B) A, B, C and D only
(C) A, C and D only
(D) A, B, D and E only

Open solution

Q13. NEET 2023 Manipur

The de Broglie wavelength associated with an electron, accelerated by a potential difference of 81 V is given by:

(A) 13.6 nm
(B) 136 nm
(C) 1.36 nm
(D) 0.136 nm

Open solution

Q14. NEET 2023 Manipur

The maximum kinetic energy of the emitted photoelectrons in photoelectric effect is independent of:

(A) work function of material
(B) intensity of incident radiation
(C) frequency of incident radiation
(D) wavelength of incident radiation

Open solution

Q15. NEET 2023

The work functions of Caesium (Cs), potassium (K) and Sodium (Na) are $2.14\ \mathrm{eV}, 2.30\ \mathrm{eV}$ and $2.75\ \mathrm{eV}$ respectively. If incident electromagnetic radiation has an incident energy of $2.20\ \mathrm{eV}$, which of these photosensitive surfaces may emit photoelectrons?

(A) Both Na and K
(B) K only
(C) Na only
(D) Cs only

Open solution

Q16. NEET 2023

The minimum wavelength of X-rays produced by an electron accelerated through a potential difference of V volts is proportional to :

(A) $\frac{1}{V}$
(B) $\frac{1}{\sqrt{V}}$
(C) V^2
(D) $\sqrt{V}$

Open solution

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Q17. NEET 2022 Phase 2

The light rays having photons of energy 4.2 eV are falling on a metal surface having a work function of 2.2 eV. The stopping potential of the surface is

(A) 6.4 V (B) 2 eV (C) 2 V (D) 1.1 V

Open solution

Q18. NEET 2022 Phase 2

The threshold frequency of a photoelectric metal is v_0. If light of frequency $4v_0$ is incident on this metal, then the maximum kinetic energy of emitted electrons will be :

(A) $4 hv_0$ (B) hv_0 (C) $2 hv_0$ (D) $3 hv_0$

Open solution

Q19. NEET 2022 Phase 1

When two monochromatic lights of frequency, v and $\frac{v}{2}$ are incident on a photoelectric metal, their stopping potential becomes $\frac{V_s}{2}$ and V_s respectively. The threshold frequency for this metal is

(A) 2v (B) 3v (C) $\frac{2}{3}$v (D) $\frac{3}{2}$v

Open solution

Q20. NEET 2021

The number of photons per second on an average emitted by the source of monochromatic light of wavelength 600nm, when it delivers the power of 3.3×10^{-3} watt will be : (h = 6.6×10^{-34} Js)

(A) 10^{15} (B) 10^{18} (C) 10^{17} (D) 10^{16}

Open solution

Q21. NEET 2021

An electromagnetic wave of wavelength 'λ' is incident on a photosensitive surface of negligible work function. If 'm' mass is of photoelectron emitted from the surface has de-Broglie wavelength λ_d, then :

(A) $\lambda = \left(\frac{2h}{mc}\right)\lambda_d{}^2$ (B) $\lambda = \left(\frac{2m}{hc}\right)\lambda_d{}^2$

(C) $\lambda_d = \left(\frac{2mc}{h}\right)\lambda^2$ (D) $\lambda = \left(\frac{2mc}{h}\right)\lambda_d{}^2$

Open solution

Q22. NEET 2020 Phase 1

Light of frequency 1.5 times the threshold frequency is incident on a photosensitive material. What will be the photoelectric current if the frequency is halved and intensity is doubled?

(A) four times (B) one-fourth (C) zero (D) doubled

Open solution

Q23. NEET 2020 Phase 1

Light with an average flux of 20 W/cm^2 falls on a non-reflecting surface at normal incidence having surface area 20 cm^2. The energy received by the surface during time span of 1 minute is :

(A) 12×10^3 J (B) 24×10^3 J (C) 48×10^3 J (D) 10×10^3 J

Open solution

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Q24. NEET 2020 Phase 1

An electron is accelerated from rest through a potential difference of V volt. If the de-Broglie wavelength of the electron is 1.227 $\times$ 10^{-2} nm, the potential difference is :

(A) 10^2 V (B) 10^3 V (C) 10^4 V (D) 10 V

Open solution

Q25. NEET 2019

An electron is accelerated through a potential difference of 10,000 V. Its de Broglie wavelength is (nearly) : (m_e = 9 $\times$ 10^{-31} kg)

(A) 12.2 × 10^{-12} m (B) 12.2 × 10^{-13} m (C) 12.2 × 10^{-14} m (D) 12.2 nm

Open solution

Q26. NEET 2018

An electron of mass m with an initial velocity $\vec{v} = v_0\hat{i}$ ($v_0 > 0$) enters an electric field $\vec{E} = -\vec{E_0}\hat{i}$ (E_0 = constant > 0) at t = 0. If λ_0 is its de-Broglie wavelength initially, then its de- Broglie wavelength at time t is

(A) $\frac{\lambda_0}{\left(1+\frac{eE_0}{mv_0}t\right)}$ (B) $\lambda_0\left(1+\frac{eE_0}{mv_0}t\right)$

(C) $\lambda_0 t$ (D) λ_0

Open solution

Q27. NEET 2018

When the light of frequency $2\upsilon_0$ (where υ_0 is threshold frequency), is incident on a metal plate, the maximum velocity of electrons emitted is v_1 . When the frequency of the incident radiation is increased to $5\upsilon_0$, the maximum velocity of electrons emitted from the same plate is v_2 . The ratio of v_1 to v_2 is

(A) 1 : 2 (B) 1 : 4 (C) 4 : 1 (D) 2 : 1

Open solution

Q28. NEET 2017

The de-Broglie wavelength of a neutron in thermal equilibrium with heavy water at a temperature T (kelvin) and mass m, is

(A) $\frac{h}{\sqrt{3mkT}}$ (B) $\frac{2h}{\sqrt{3mkT}}$ (C) $\frac{2h}{\sqrt{mkT}}$ (D) $\frac{h}{\sqrt{mkT}}$

Open solution

Q29. NEET 2016 Phase 2

Electrons of mass m with de-Broglie wavelength λ fall on the target in an X-ray tube. The cutoff wavelength (λ_0) of the emitted X-ray is

(A) $\lambda_0 = \frac{2mc\lambda^2}{h}$ (B) $\lambda_0 = \frac{2h}{mc}$

(C) $\lambda_0 = \frac{2m^2c^2\lambda^3}{h^2}$ (D) $\lambda_0 = \lambda$

Open solution

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Q30. NEET 2016 Phase 2

Photons with energy 5 eV are incident on a cathode C in a photoelectric cell. The maximum energy of emitted photoelectrons is 2 eV. When photons of energy 6 eV are incident on C, no photoelectrons will reach the anode A, if the stopping potential of A relative to C is

Open solution

(A) + 3 V (B) +4 V (C) -1 V (D) -3 V

Q31. NEET 2016 Phase 1

An electron of mass m and a photon have same energy E. The ratio of de-Broglie wavelengths associated with them is

Open solution

(A) $c(2mE)^{\frac{1}{2}}$ (B) $\frac{1}{c}\left(\frac{2m}{E}\right)^{\frac{1}{2}}$

(C) $\frac{1}{c}\left(\frac{E}{2m}\right)^{\frac{1}{2}}$ (D) $\left(\frac{E}{2m}\right)^{\frac{1}{2}}$

Q32. NEET 2016 Phase 1

When a metallic surface is illuminated with radiation of wavelength λ, the stopping potential is V. If the same surface is illuminated with radiation of wavelength 2 λ, the stopping potential is $\frac{V}{4}$. The threshold wavelength for the metallic surface is

Open solution

(A) $\frac{5}{2}\lambda$ (B) 3λ (C) 4λ (D) 5λ

Q33. AIPMT 2015

A photoelectric surface is illuminated successively by monochromatic light of wavelength λ and $\lambda/2$. If the maximum kinetic energy of the emitted photoelectrons in the second case is 3 times that in the first case, the work function of the surface of the material is
(h = Planck's constant, C = speed of light)

Open solution

(A) $\frac{2hc}{\lambda}$ (B) $\frac{hc}{3\lambda}$ (C) $\frac{hc}{2\lambda}$ (D) $\frac{hc}{\lambda}$

Q34. AIPMT 2015

Light of wavelength 500 nm is incident on a metal with work function 2.28 eV. The de Broglie wavelength of the emitted electron is

Open solution

(A) $\geq 2.8 \times 10^{-9}$ m (B) $\leq 2.8 \times 10^{-12}$ m

(C) $< 2.8 \times 10^{-10}$ m (D) $< 2.8 \times 10^{-9}$ m

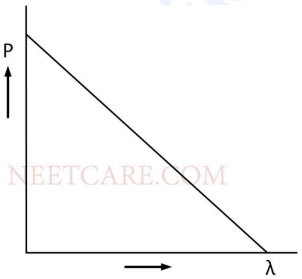
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Q35. AIPMT 2015 Cancelled Paper

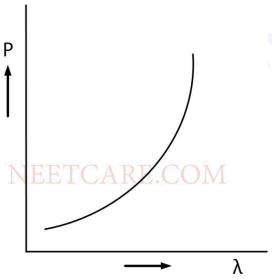
Open solution

Which of the following figures represent the variation of particle momentum and the associated de-Broglie wavelength?

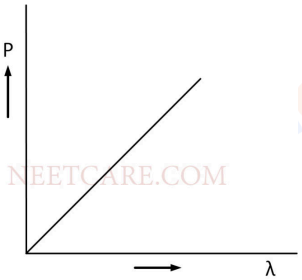
(A)



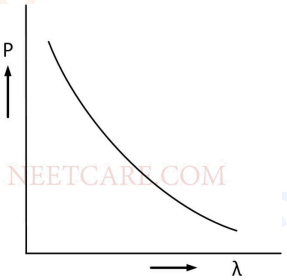
(B)



(C)



(D)



Q36. AIPMT 2015 Cancelled Paper

Open solution

A certain metallic surface is illuminated with monochromatic light of wavelength, λ. The stopping potential for photo-electric current for this light is $3V_0$. If the same surface is illuminated with light of wavelength 2λ, the stopping potential is V_0. The threshold wavelength for this surface for photo-electric effect is

(A) $\frac{\lambda}{4}$ (B) $\frac{\lambda}{6}$ (C) 6λ (D) 4λ

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Q37. AIPMT 2014

When the energy of the incident radiation is increased nby 20%, the kinetic energy of the photoelectrons emitted from a metal surface increased from 0.5 eV to 0.8 eV. The work function of the metal is

Open solution

(A) 0.65 eV (B) 1.0 eV (C) 1.3 eV (D) 1.5 eV

Q38. AIPMT 2014

If the kinetic energy of the particle is increased to 16 times its previous value, the percentage change in the de Broglie wavelength of the particle is

Open solution

(A) 25 (B) 75 (C) 60 (D) 50

Q39. NEET 2013 (Karnataka)

A source of light is placed at a distance of 50 cm from a photo cell and the stopping potential is found to be V_0. If the distance between the light source and photo cell is made 25 cm, the new stopping potential will be :

Open solution

(A) $V_0/2$ (B) V_0 (C) $4V_0$ (D) $2V_0$

Q40. NEET 2013 (Karnataka)

The de-broglie wavelength of neutrons in thermal equilibrium at temperature T is

Open solution

(A) $\frac{3.08}{\sqrt{T}}\mathring{A}$ (B) $\frac{0.308}{\sqrt{T}}\mathring{A}$

(C) $\frac{0.0308}{\sqrt{T}}\mathring{A}$ (D) $\frac{30.8}{\sqrt{T}}\mathring{A}$

Q41. NEET 2013

A parallel beam of fast moving electrons is incident normally on a narrow slit. A fluorescent screen is placed at a large distance from the slit. If the speed of the electrons is increased, which of the following statements is correct ?

Open solution

(A) The angular width of the central maximum will decrease.

(B) The angular wifth of the central maximum will be unaffected.

(C) Diffraction pattern is not observed on the screen in the case of electrons.

(D) The angular width of the central maximum of the diffraction pattern will increase.

Q42. NEET 2013

The wavelength λ_e of an electron and λ_p of a photon of same energy E are related by

Open solution

(A) $\lambda_p \propto \sqrt{\lambda_e}$ (B) $\lambda_p \propto \frac{1}{\sqrt{\lambda_e}}$

(C) $\lambda_p \propto {\lambda_e}^2$ (D) $\lambda_p \propto \lambda_e$

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Q43. NEET 2013

For photoelectric emission from certain metal the cutoff frequency is υ. If radiation of frequency 2υ impinges on the metal plate, the maximum possible velocity of the emitted electron will be (m is the electron mass)

(A) $\sqrt{\frac{2h\upsilon}{m}}$ (B) $2\sqrt{\frac{h\upsilon}{m}}$

(C) $\sqrt{\frac{h\upsilon}{(2m)}}$ (D) $\sqrt{\frac{h\upsilon}{m}}$

Open solution

Q44. AIPMT 2012 Mains

If the momentum of an electron is changed by P, then the de Broglie wavelength associated with changes by 0.5%. The initial momentum of electron will be

(A) 200 P (B) 400 P (C) $\frac{P}{200}$ (D) 100 P

Open solution

Q45. AIPMT 2012 Mains

Two radiations of photons energies 1 eV and 2.5 eV, successively illuminate a photosensitive metallic surface of work function 0.5 eV. The ratio of the maximum speeds of the emitted electrons is

(A) 1 : 4 (B) 1 : 2 (C) 1 : 1 (D) 1 : 5

Open solution

Q46. AIPMT 2012 Prelims

A 200 W sodium street lamp emits yellow light of wavelength 0.6 μm. Assuming it to be 25% efficient in converting electrical energy to light, the number of photons of yellow light it emits per second is

(A) 1.5×10^{20} (B) 6×10^{18} (C) 62×10^{20} (D) 3×10^{19}

Open solution

Q47. AIPMT 2012 Prelims

Monochromatic radiation emitted when electron on hydrogen atom jumps from first excited to the ground state irradiates a photosensitive material. The stopping potential is measured to be 3.57 V. The threshold frequency of the material is

(A) 4×10^{15} Hz (B) 5×10^{15} Hz (C) 1.6×10^{15} Hz (D) 2.5×10^{15} Hz

Open solution

Q48. AIPMT 2012 Prelims

An α-particle moves in a circular path of radius 0.83 cm in the presence of a magnetic field of 0.25 Wb/m^2. The de Broglie wavelength associated with the particle will be

(A) 1 $\mathring{A}$ (B) 0.1 $\mathring{A}$

(C) 10 $\mathring{A}$ (D) 0.01 $\mathring{A}$

Open solution

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Q49. AIPMT 2011 Mains

The threshold frequency for a photosensitive metal is 3.3×10^{14} Hz. If light of frequency 8.2×10^{14} Hz is incident on this metal, the cut- off voltage for the photoelectron emission is nearly

(A) 1 V (B) 2 V (C) 3 V (D) 5 V

Open solution

Q50. AIPMT 2011 Prelims

Photoelectric emission occurs only when the incident light has more than a certain minimum

(A) power (B) wavelength (C) intensity (D) frequency

Open solution

Q51. AIPMT 2011 Prelims

Light of two different frequencies whose photons have energies 1 eV and 2.5 eV respectively illuminate a metallic surface whose work function is 0.5 eV successively. Ratio of maximum speeds of emitted electrons will be

(A) 1 : 4 (B) 1 : 2 (C) 1 : 1 (D) 1 : 5

Open solution

Q52. AIPMT 2011 Prelims

Electrons used in an electron microscope are accelerated by a voltage of 25 kV. If the voltage is increased to 100 kV then the de-Broglie wavelength associated with the electrons would

(A) increase by 2 times (B) decrease by 2 times

(C) decrease by 4 times (D) increase by 4 times

Open solution

Q53. AIPMT 2011 Prelims

In photoelectric emission process from a metal of work function 1.8 eV, the kinetic energy of most energetic electrons is 0.5 eV. The corresponding stopping potential is

(A) 1.8 V (B) 1.3 V (C) 0.5 V (D) 2.3 V

Open solution

Q54. AIPMT 2010 Mains

When monochromatic radiation of intensity I falls on a metal surface, the number of photoelectrons and their maximum kinetic energy are N and T respectively. If the intensity of radiation is $2I$, the number of emitted electrons and their maximum kinetic energy are respectively

(A) N and 2T (B) 2N and T (C) 2N and 2T (D) N and T

Open solution

Q55. AIPMT 2010 Mains

The electron in the hydrogen atom jumps from excited state (n = 3) to its ground state (n = 1) and the photons thus emitted irradiate a photosensitive material. If the work function of the material is 5.1 eV, the stopping potential is estimated to be (the energy of the electron in n^{th} state $E_n = \frac{-13.6}{n^2} eV$)

(A) 5.1 V (B) 12.1 V (C) 17.2 V (D) 7 V

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Q56. AIPMT 2010 Prelims

A beam of cathode rays is subjected to crossed electric (E) and magnetic fields (B). The fields are adjusted such that the beam is not deflected. The specific charge of the cathode rays is given by

(Where V is the potential difference between cathode and anode)

(A) $\frac{B^2}{2VE^2}$ (B) $\frac{2VB^2}{E^2}$ (C) $\frac{2VE^2}{B^2}$ (D) $\frac{E^2}{2VB^2}$

Open solution

Q57. AIPMT 2010 Prelims

A source S_1 is producing, 10^{15} photons per second of wavelength 5000 $\mathring{A}$. Another source S_2 is producing 1.02×10^{15} photons per second of wavelength 5100 $\mathring{A}$. Then, (power of S_2)/(power of S_1) is equal to

(A) 1.00 (B) 1.02 (C) 1.04 (D) 0.98

Open solution

Q58. AIPMT 2010 Prelims

The potential difference that must be applied to stop the fastest photoelectrons emitted by a nickel surface, having work functions 5.01 eV, when ultraviolet light of 200 nm falls on it, must be

(A) 2.4 V (B) $-$ 1.2 V (C) $-$ 2.4 V (D) 1.2 V

Open solution

Q59. AIPMT 2009

The number of photo electrons emitted for light of a frequency v (higher than the threshold frequency v_0) is proportional to

(A) threshold frequency (v_0) (B) intensity of light

(C) frequency of light (v_0) (D) $v - v_0$

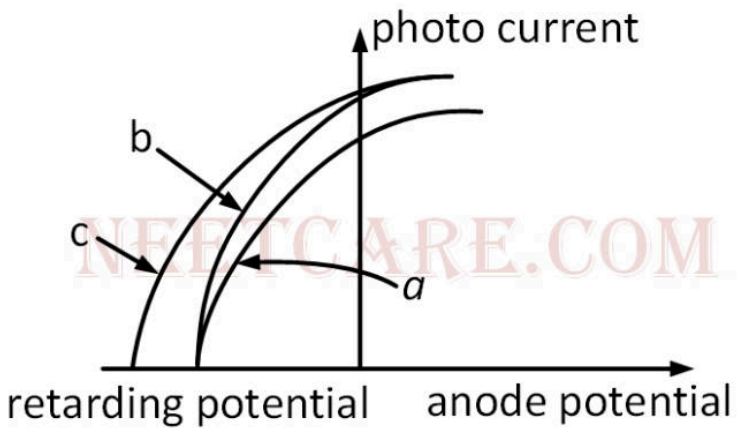
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Q60. AIPMT 2009

The figure shows a plot of photo current versus anode potential for a photo sensitive surface for three different radiations. Which one of the following is a correct statement?

Open solution



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(A) Curves (a) and (b) represent incident radiations of same frequency but of different intensities.

(B) Curves (b) and (c) represent incident radiations of different frquencies and different intensities.

(C) Curves (b) and (C) represent incident radiations of same frequency having same intensity.

(D) Curves (a) and (b) represent incident radiations of different frequencies and different intensities

Q61. AIPMT 2009

Monochromatic light of wavelength 667 nm is produced by a helium neon laser. The power emitted is 9 mW. The number of photons arriving per sec. on the average at a target irradiated by this beam is

Open solution

(A) 3×10^{16} (B) 9×10^{15} (C) 3×10^{19} (D) 9×10^{17}

Q62. AIPMT 2008

The work function of a surface of a photosensitive material is 6.2 eV. The wavelength of the incident radiation for which the stopping potential is 5 V lies in the

Open solution

(A) infrared region (B) X-ray region

(C) Ultraviolet region (D) Visible region

Q63. AIPMT 2008

A particle of mass 1 mg has the same wavelength as an electron moving with a velocity of 3×10^6 m s^{-1}. The velocity of the particle is

Open solution

(A) 3×10^{-31} ms^{-1} (B) 2.7×10^{-21} ms^{-1}

(C) 2.7×10^{-18} ms^{-1} (D) 9×10^{-2} ms^{-1}

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Q64. AIPMT 2008

In the phenomenon of electric discharge through gases at low pressure, the coloured glow in the tube appears as a result of

(A) collisions between the charged particles emitted from the cathode and the atoms of the gas

(B) collision between different electrons of the atoms of the gas

(C) excitation of electrons in the atoms

(D) collision between the atoms of the gas

Open solution

Q65. AIPMT 2007

A beam of electron passes undeflected through mutually perpendicular electric and magnetic fields. If the electric field is switched off, and the same magnetic field is maintained, the electrons move

(A) in a circular orbit

(B) along a parabolic path

(C) along a straight line

(D) in an elliptical orbit.

Open solution

Q66. AIPMT 2007

Monochromatic light of frequency 6.0×10^{14} Hz is produced by a laser. The power emitted is 2×10^{-3} W. The number of photons emitted, on the average, by the source per second is

(A) 5×10^{16} (B) 5×10^{17} (C) 5×10^{14} (D) 5×10^{15}

Open solution

Q67. AIPMT 2007

A 5 watt source emits monochromatic light of wavelength 5000 $\mathring{A}$. When placed 0.5 m away, it liberates photoelectrons from a photosensitive metallic surface. When the source is moved to a distance of 1.0 m, the number of photoelectrons liberated will be reduced by a factor of

(A) 8 (B) 16 (C) 2 (D) 4

Open solution

Q68. AIPMT 2006

A photocell employs photoelectric effect to convert

(A) change in the frequency of light into a change in the electric current

(B) change in the frequency of light into a change in electric voltage

(C) change in the intensity of illumination into a change in photoelectric current

(D) change in the intensity of illumination into a change in the work function of the photocathode.

Open solution

Q69. AIPMT 2006

When photons of energy $h\upsilon$ fall on an aluminimum plate (of work function E_0), photoelectrons of maximum kinetic energy K are ejected. If the frequency of radiation is doubled, the maximum kinetic energy of the ejected photoelectrons will be

(A) $K + h\upsilon$ (B) $K + E_0$ (C) 2K (D) K.

Open solution

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Q70. AIPMT 2006

In a discharge tube ionization of enclosed gas is produced due to collisions between

(A) neutral gas atoms/molecules

(B) positive ions and neutral atoms/molecules

(C) negative electrons and neutral atoms/molecules

(D) photons and neutral atoms/molecules.

Open solution

Q71. AIPMT 2006

The momentum of a photon of energy 1 MeV in kg m/s will be

(A) 5×10^{-22} (B) 0.33×10^{6} (C) 7×10^{-24} (D) 10^{-22}

Open solution

Q72. AIPMT 2005

The work functions for metals A, B and C are respectively 1.92 eV, 2.0 eV and 5 eV. According to Einstein's equation the metals which will emit photoelectrons for a radiation of wavelength 4100 $\mathring{A}$ is/are

(A) A only

(B) A and B only

(C) all the three metals

(D) none

Open solution

Q73. AIPMT 2005

A photosensitive metallic surface has work function, $h\upsilon_0$. If photons of energy $2h\upsilon_0$ fall on this surface, the electrons come out with a maximum velocity of 4×10^6 m/s. When the photon energy is increased to 5 $h\upsilon_0$, then maximum velocity of photoelectrons will be

(A) 2×10^7 m/s (B) 2×10^6 m/s (C) 8×10^6 m/s (D) 8×10^5 m/s

Open solution

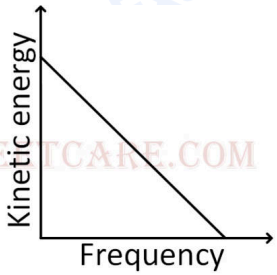
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Q74. AIPMT 2004

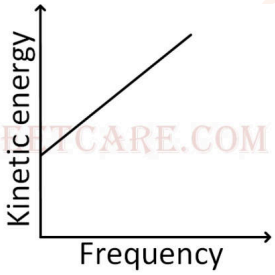
Open solution

According to Einstein's photoelectric equation, the graph between the kinetic energy of photoelectrons ejected and the frequency of incident radiation is

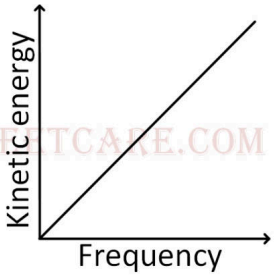
(A)



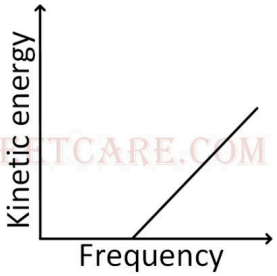
(B)



(C)



(D)



Q75. AIPMT 2003

Open solution

A photoelectric cell is illuminated by a point source of light 1 m away. When the source is shifted to 2 m then

(A) each emitted electron carries one quarter of the initial energy

(B) number of electrons emitted is half the initial number

(C) each emitted electron carries half the initial energy

(D) number of electrons emitted is a quarter of the initial number

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Q76. AIPMT 2003

J.J. Thomson's cathode-ray tube experiment demonstrated that

(A) cathode rays are streams of negatively charged ions

(B) all the mass of an atom is essentially in the nucleus

(C) the e/m of electrons is much greater than the e/m of protons

(D) the e/m ratio of the cathode-ray particles changes when a different gas is placed in the discharge tube

Open solution

Q77. AIPMT 2002

The value of Planck's constant is

(A) 6.63×10^{-34} J/sec.

(B) 6.63×10^{-34} kg-m^2/sec

(C) 6.63×10^{-34} kg-m^2

(D) 6.63×10^{-34} J-sec.

Open solution

Q78. AIPMT 2002

If particles are moving with same velocity, then which has maximum de Broglie wavelength?

(A) proton (B) α-particle (C) neutron (D) β-particle

Open solution

Q79. AIPMT 2002

When ultraviolet rays incident on metal plate then photoelectric effect does not occur, it occurs by incidence of

(A) infrared rays (B) X-rays (C) radio wave (D) micro wave

Open solution

Q80. AIPMT 2002

Which of the following is not the property of cathode rays ?

(A) In produces heating effect

(B) It does not deflect in electric field

(C) It casts shadow

(D) It produces fluorscence.

Open solution

Q81. AIPMT 2001

Which one among the following shows particle nature of light?

(A) photo electric effect

(B) interference

(C) refraction

(D) polarization.

Open solution

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Q82. AIPMT 2001

In Thomson mass spectrograph $\vec{E} \perp \vec{B}$ then the velocity of electron beam will be

(A) $\frac{|\vec{E}|}{|\vec{B}|}$

(B) $\vec{E} \times \vec{B}$

(C) $\frac{|\vec{B}|}{|\vec{E}|}$

(D) $\frac{\vec{E}^2}{\vec{B}^2}$

Open solution

Q83. AIPMT 2001

A photo-cell is illuminated by a source of light, which is placed at a distance d from the cell. If the distance become d/2, then number of electrons emitted per second, will be

(A) remain same (B) four times (C) two times (D) one-fourth

Open solution

Q84. AIPMT 2000

By photoelectric effect, Einstein proved

(A) E = $h\upsilon$ (B) K.E. = $\frac{1}{2}$ mv^2

(C) E = mc^2 (D) E = $\frac{-Rhc^2}{n^2}$

Open solution

Q85. AIPMT 2000

Who evaluated the mass of electron indirectly with help of charge

(A) Thomson (B) Millikan (C) Rutherford (D) Newton

Open solution

MCQ (More than One Correct Answer) 1 questions

Q86. NEET 2017

The photoelectric threshold wavelength of silver is 3250×10^{-10} m. The velocity of the electron ejected from a silver surface by ultraviolet light of wavelength 2536×10^{-10} m is {Given h = 4.14×10^{-15} eV s and c = 3×10^8 m s^{-1}]

(A) $\approx 0.6 \times 10^6$ m s^{-1} (B) $\approx 61 \times 10^3$ m s^{-1}

(C) $\approx 0.3 \times 10^6$ m s^{-1} (D) $\approx 6 \times 10^5$ m s^{-1}

Open solution

Out of syllabus 1 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

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Q87. AIPMT 2011 Prelims · **MCQ (Single Correct Answer)**

In the Davisson and Germer experiment, the velocity of electrons emitted from the electron gun can be increased by

Open solution

(A) increasing the potential difference between the anode and filament

(B) increasing the filament current

(C) decreasing the filament current

(D) decreasing the potential difference between the anode and flament

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Answer Key

MCQ (Single Correct Answer)

1. **D**	2. **B**	3. **B**	4. **D**	5. **C**
6. **D**	7. **C**	8. **A**	9. **A**	10. **C**
11. **D**	12. **B**	13. **D**	14. **B**	15. **D**
16. **A**	17. **C**	18. **D**	19. **D**	20. **D**
21. **D**	22. **C**	23. **B**	24. **C**	25. **A**
26. **A**	27. **A**	28. **A**	29. **A**	30. **D**
31. **C**	32. **B**	33. **C**	34. **A**	35. **D**
36. **D**	37. **B**	38. **B**	39. **B**	40. **D**
41. **A**	42. **C**	43. **A**	44. **A**	45. **B**
46. **A**	47. **C**	48. **D**	49. **B**	50. **D**
51. **B**	52. **B**	53. **C**	54. **B**	55. **D**
56. **D**	57. **A**	58. **B**	59. **B**	60. **A**
61. **A**	62. **C**	63. **C**	64. **A**	65. **A**
66. **D**	67. **D**	68. **C**	69. **A**	70. **C**
71. **A**	72. **B**	73. **C**	74. **D**	75. **D**
76. **C**	77. **D**	78. **D**	79. **B**	80. **B**
81. **A**	82. **A**	83. **B**	84. **A**	85. **A**

MCQ (More than One Correct Answer)

86. **A, D**

Out of syllabus

87. **A**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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